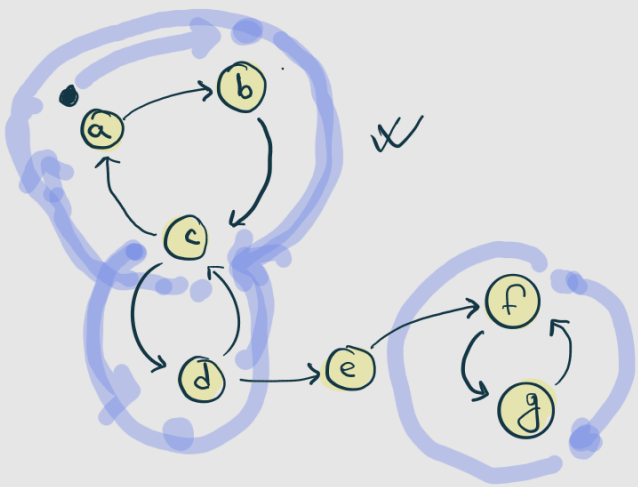


Strongly Connected Component



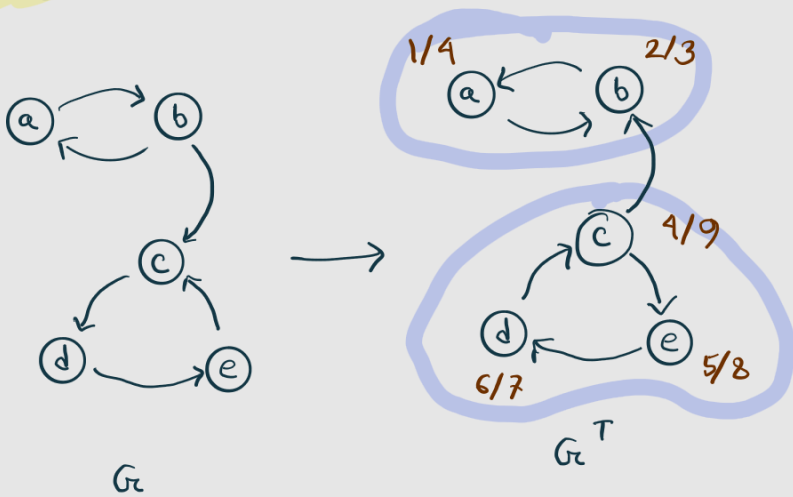
$\{a, b, c\}$
 $\{c, d\}$

$\{f, g\}$
cycle

$\{a, b, c, d\}$

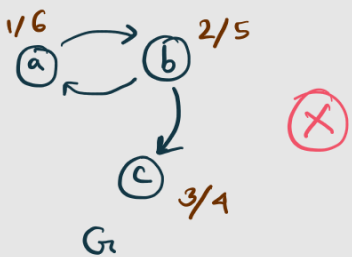
$\{f, g\}$
strongly connected components

Kosaraju Algorithm : Ending time

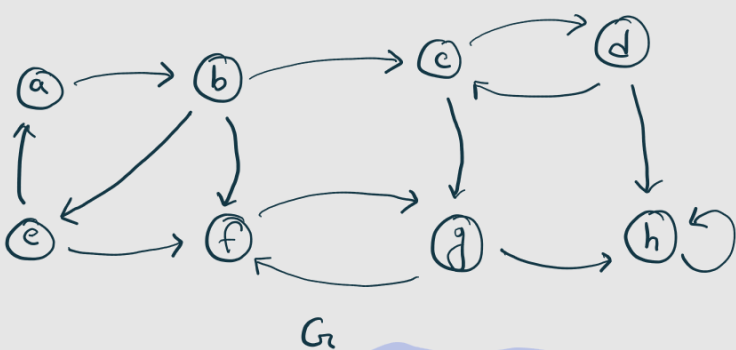
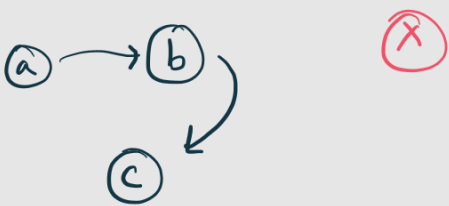


$\{a, b\}$
 $\{c, d, e\}$

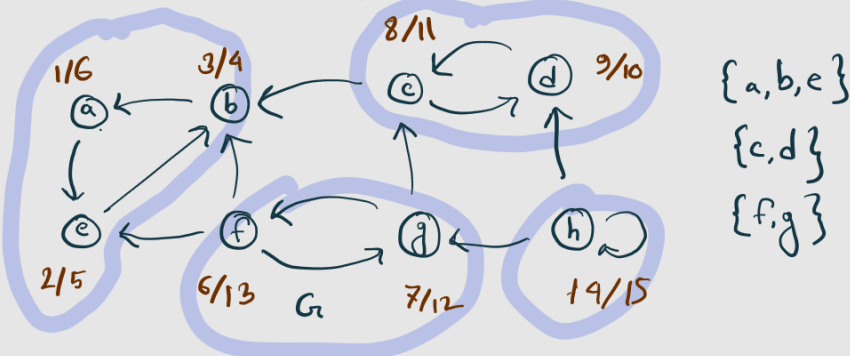
Is SCC possible without G^T ?



Is SCC possible in DAG (Directed Acyclic Graph)?



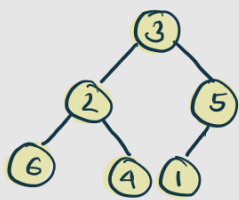
Ans:



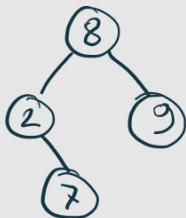
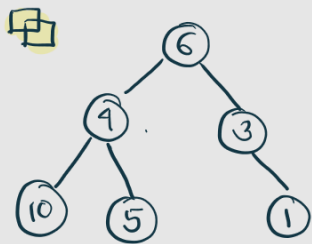
Scenario: ✓ Social Network Circle / Mutuals [Reachability]
✓ Crime connections

DSU [Disjoint set Union] ✓

* Almost always undirected



0	1	2	3	4	5	6
-1	5	3	3	2	3	2



0	1	2	3	4	5	6	7	8	9	10
-1	3	8	6	6	4	6	2	8	8	4

(a) How many trees are here? [Ans: 2]

(b) Do (5, 7) belong in the same tree?

No

5 → 4 → 6 (Root)

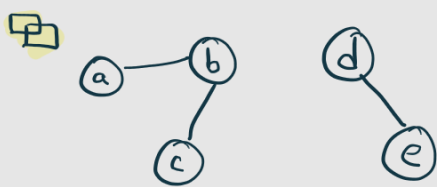
7 → 2 → 8 (Root)

(c) Do (7, 9) belong in the same tree?

Yes

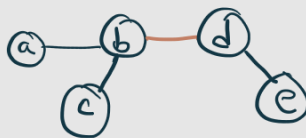
7 → 2 → 8 (Root)

9 → 8 (Root)



Make c and d friends

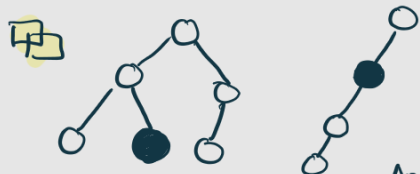
To avoid unbalance



b	b	b	d	d
a	b	c	d	e

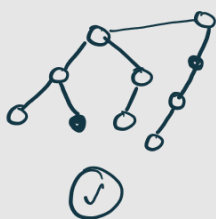
* Find Parent

* Union



Make black nodes friends

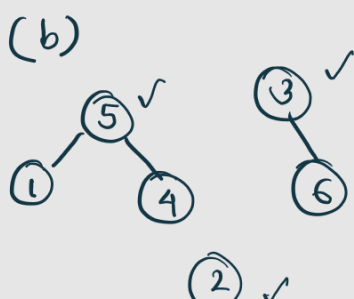
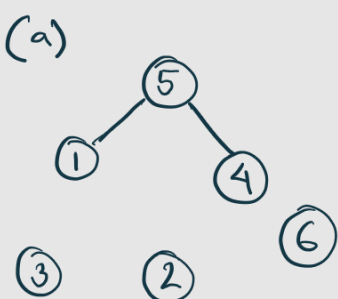
Ans:



(a) Make 1, 5 and 4 friends

(b) Make 3 and 6 friends

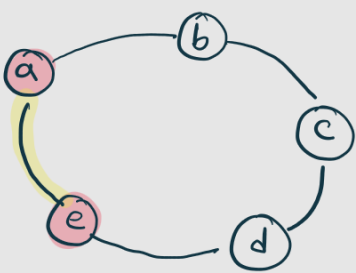
(c) How many trees are here



(c) 3

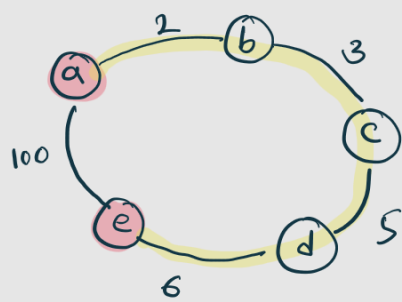
Scenario: Same group

Shortest Path



Shortest Path from
a to e

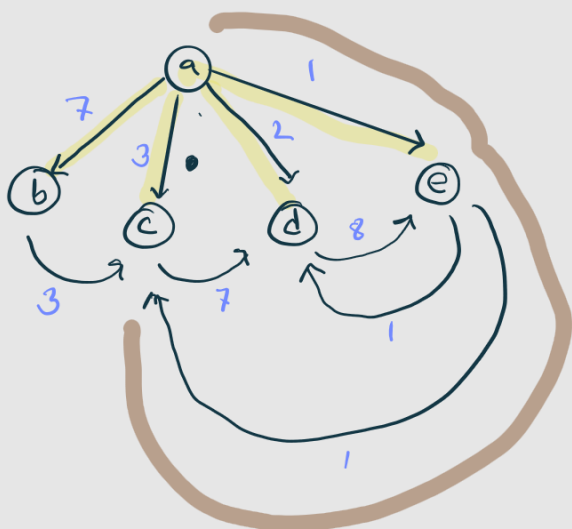
(BFS) → Unweighted
✓ shortest Path



Shortest Path from
a to e

[Use Dijkstra Algorithm]

(DFS) → weighted shortest
Path



Show Dijkstra Simulation

a - 0

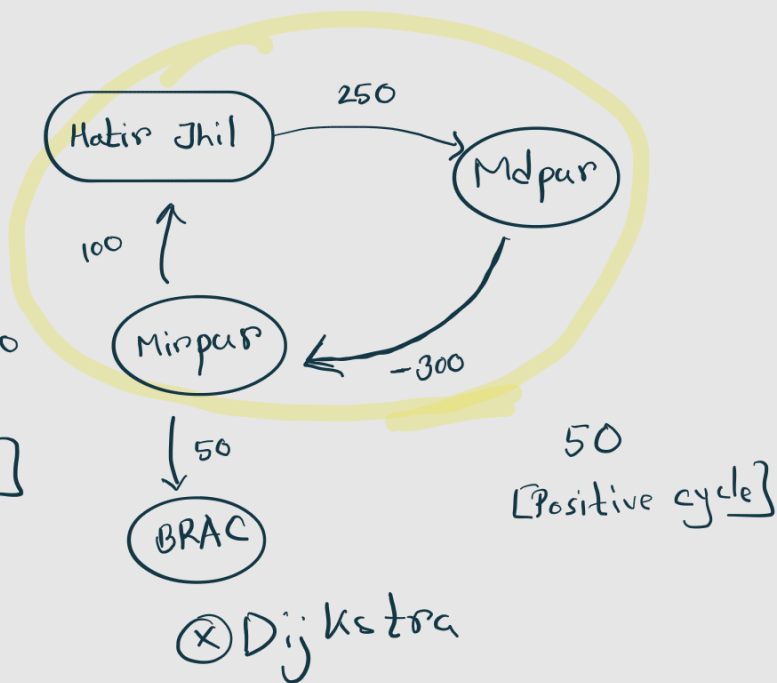
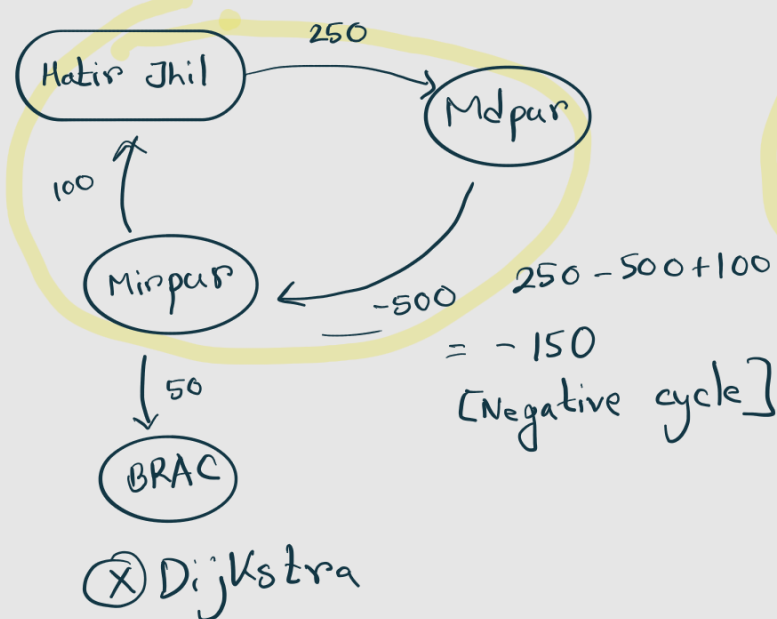
b - ∞ 7

c - ∞ 2 [Relaxation]

d - ∞ 2

e - ∞ 1

- Type:
1. Single source shortest Path (SSSP)
 2. Single destination
 3. Single Pair
 4. All pair shortest Path



Limitation: Non negative weight needed

Scenario: ✓ Google Maps / Flight cost

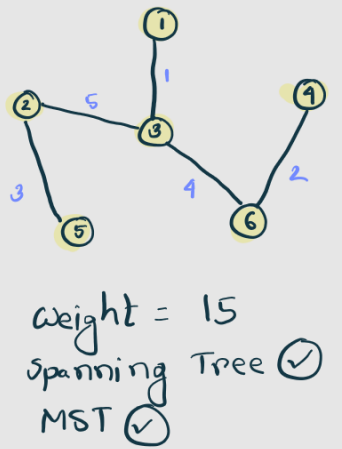
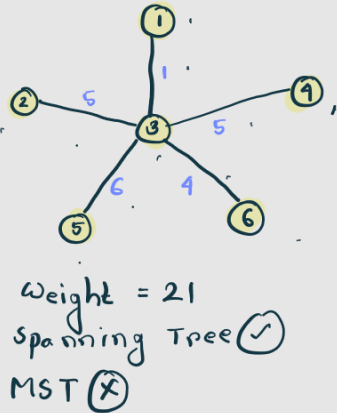
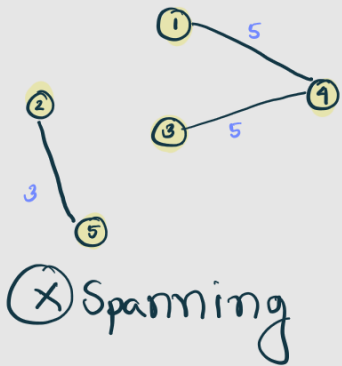
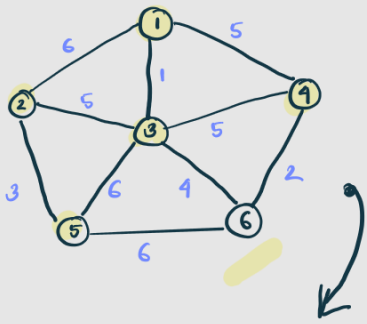
Minimum Spanning Tree (MST)

✓ ① All node

✓ ② Edge = $n-1$

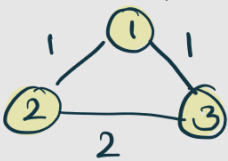
③ Minimum weight

Find MST

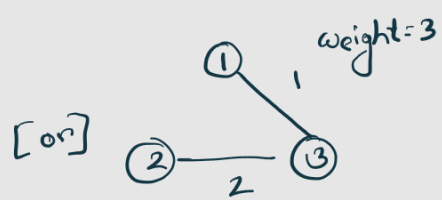
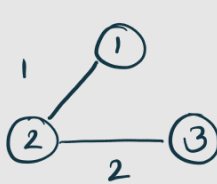


Can minimum spanning tree always be the same?

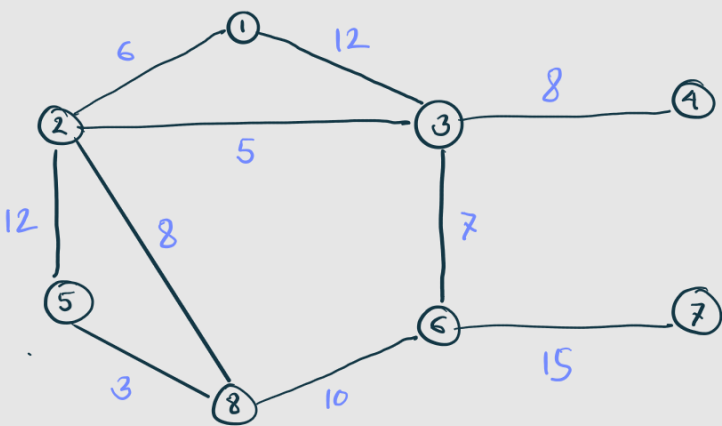
Ans: No



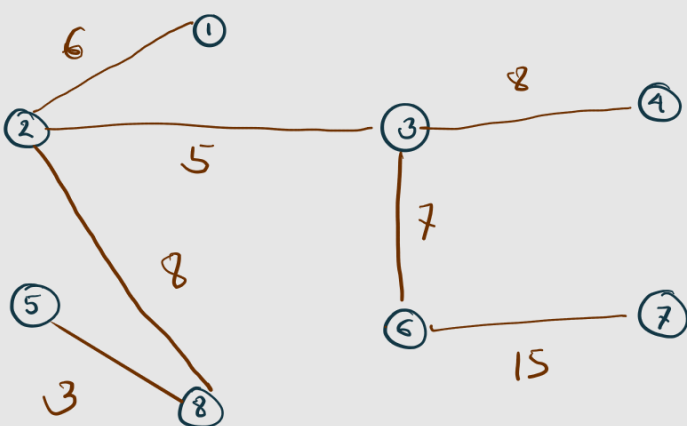
MST:



Find MST using **Kruskal Algorithm** and show **Minimum cost**



$w(u,v)$	(u,v)
3	(5,8) ✓
5	(2,3) ✓
6	(1,2) ✓
7	(3,6) ✓
8	(2,8) ✓
8	(3,4) ✓
10	(6,8) (X)
12	(1,3) (X)
12	(2,5) (X)
15	(6,7)



Minimum cost = 53 [v.v.g]

Scenario: ✓ Minimum cost to make road
✓ Oil pipeline / Cables

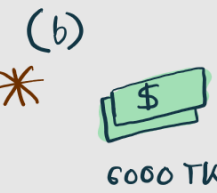
Knapsack

① Fractional Knapsack:



চাল — 50 kg — 2500 TK	$\frac{2500}{50} = 50 \text{ TK/kg}$
ডাল — 20 kg — 4000 TK	200 tk/kg ✓
চিনি — 100 kg — 10 000 TK	100 tk/kg
গোলাব চাল — 50 kg — 5000 TK	100 tk/kg =

ডাল (20 kg) + গোলাব চাল (10 kg)



চাল — 50 kg — 2500 TK	$\frac{50}{2500} = 0.02 \text{ kg/TK}$
ডাল — 20 kg — 4000 TK	$5 \times 10^{-3} \text{ kg/TK}$
চিনি — 100 kg — 10 000 TK	$\frac{100}{10000} = 0.01 \text{ kg/TK}$
গোলাব চাল — 50 kg — 5000 TK	100 kg/TK

গোলাব চাল (5000) + চাল (1000)

② Zero One Knapsack

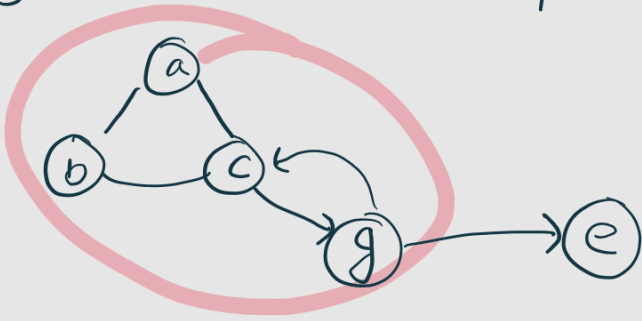


weight	Profit
5 kg	20 ✓
2 kg ✓	30
4 kg	15
3 kg	10

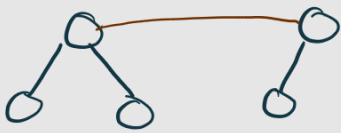
Item \ weight	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	0	20	20
2	0	0	30	30	30	30	30
3	0	0	30	30	30	30	45
4	0	0	30	30	30	40	45

Summary

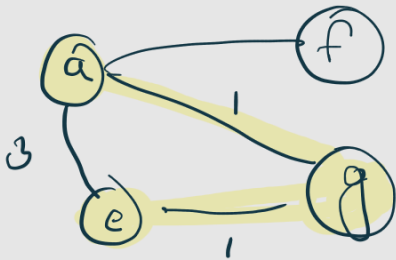
① Strongly Connected Components [Kosaraju]



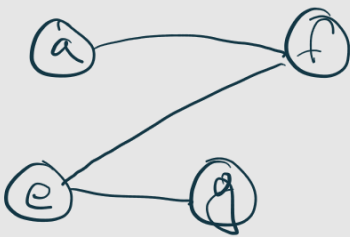
② DSU (Disjoint set Union)



③ Shortest Path (Dijkstra)



④ MST / Minimum Spanning Tree (Kruskal)



⑤ Knapsack

50 kg
100 Tk } Limitation